

Video-Based Pediatric Seizure Detection from Pose Landmark Sequences

*An ensemble of CNN-GRU and Temporal Convolutional Networks
for binary detection of infantile spasms*

Babajide Owoyele, Sharjeel Shaikh, Wim Pouw,
Moritz Schneider, Stefan Konigorski, Gerard de Melo

Hasso Plattner Institute, Universität Potsdam, Germany

2026 Video-Based Seizure Detection Challenge

Section on Computational Neurology, Charité — Universitätsmedizin Berlin

Int. Conf. on AI in Epilepsy and Other Neurological Disorders

Task — binary detection of infantile spasms (West syndrome) from 5-second segments of anonymised MediaPipe Pose: $150 \times 33 \times 5$ tensors of $x, y, z, visibility, presence$.

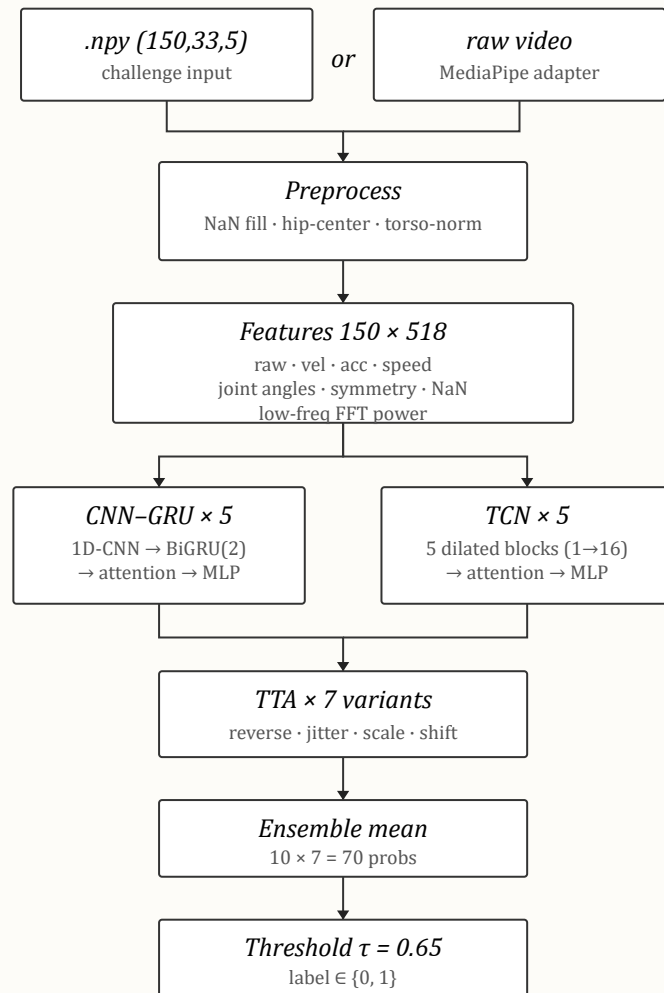
Why infantile spasms. Defining seizure type of West syndrome; onset 3–24 months; brief (1–3 s) flexor/extensor/mixed contraction, often clustered. Time-to-treatment strongly predicts cognitive outcome.

Why home video. First observed by caregivers, not clinicians. Home-video review is part of the standard diagnostic pathway.

Why pose only.

- **Privacy:** pose strips identity, room, clothing.
- **Compute:** a few hundred floats / frame, not millions.
- **Robustness:** invariant to lighting, clothing, background.

Ceiling: facial cues, vital signs and audio are *not* in the pose stream.



Two complementary backbones.

- **CNN-GRU:** recurrent inductive bias, models temporal order via hidden state.
- **Multi-scale TCN:** parallel dilated convolutions covering the full 150-frame window.

5-fold StratifiedGroupKFold grouped by patient ID — leakage-free.

7-variant TTA.

Preprocessing. Forward/back-fill NaN frames; hip-center; torso-length normalisation. Preserve binary NaN mask as a feature.

518-d per-frame features.

Component	Dim.
Raw landmarks (x,y,z,vis,pres)	165
Velocity / Acceleration / Speed	231
Joint angles (10) · Symmetry (6) · NaN	17
Low-freq FFT power (broadcast)	105
Total	518

FFT bins $\approx$ 0.2–3 Hz — clonic / myoclonic rhythm at 30 fps over a 5-s window.

Training.

- **Focal loss** $\alpha=0.75$, $\gamma=2.0$ + WeightedRandomSampler.
- **AdamW + OneCycleLR**, max_lr 5×10^{-3} , pct_start 0.3.
- Up to 120 epochs, early stop on val F_1 (patience 25).
- **Augmentation:** reverse, jitter, time-resample, frame dropout, same-class mixup.

Inference. 10 models $\times$ 7 TTA variants = 70 forward passes / sample, mean probability, threshold $\tau = 0.65$.

Same trained ensemble runs on either pre-extracted .npy or raw .mp4 via the MediaPipe adapter — no retraining.

5-fold CV on the training partition.

<i>Fold</i>	<i>CNN-GRU F_1</i>	$\hat{t}$	<i>TCN F_1</i>	$\hat{t}$
0	0.633	0.70	0.611	0.63
1	0.615	0.59	0.608	0.61
2	0.629	0.64	0.635	0.33
3	0.575	0.28	0.627	0.25
4	0.691	0.51	0.694	0.39
Mean	0.628	0.54	0.635	0.44

Per-fold thresholds span [0.25, 0.70] — different operating points on the precision–recall curve; ensemble exploits this.

Held-out leaderboard.

<i>Rank</i>	<i>Team</i>	<i>F₁</i>
1	Kramer (BU/JHU)	0.4502
2	Ramesh (NIT)	0.4286
3	Daraie (JHU)	0.4048
4	Walsh (Penn)	0.3984
5	Hazra (IIT KGP)	0.3923
⋮	⋮	⋮
19	Owoyele et al. (HPI)	0.2975

26 valid submissions, no entry exceeds F_1 0.5. Compressed leaderboard — task is genuinely hard.

Five plausible contributors, ordered roughly by expected effect size:

- **Distribution shift** between training-pool and test-pool patients. Grouped CV controls leakage but *not* pool-level shift.
- **Threshold over-fit to CV.** Per-fold $\hat{\tau}$ spans [0.25, 0.70]; the single $\tau = 0.65$ we ship is over-fit to a small CV sample.
- **F_1 brittleness** under strong class imbalance — small precision / recall shifts move F_1 a lot.
- **Train / TTA augmentation mismatch.** Aggressive training augmentation, near-identity TTA — the model is trained to tolerate perturbations the TTA does not produce.
- **Architectural mismatch.** Full-window attention pools 5 s; spasms concentrate in 1–3 s sub-windows. MIL-style sub-window pooling (used by top entries) is a better structural prior.

Concrete next actions: (i) hold out a fold for threshold calibration; (ii) refactor inference around MIL sub-window pooling.

Future work.

- **MeTRAbs** pose front-end — metric-scale 3-D, 87 joints (hand + head detail MediaPipe lacks).
- **Multi-view triangulation** via markerless dynamic extrinsic calibration — fuse synchronised views.
- **MIL sub-window pooling** matched to spasm duration.
- **MaskAnyone** integration — on-device home-video processing; only landmarks + label leave the premises.
- **Calibrated probabilities** for cost-sensitive clinical triage.

Thanks.

Section on Computational Neurology, Charité — Universitätsmedizin Berlin, and the organising committee of the 2026 Int. Conf. on AI in Epilepsy and Other Neurological Disorders, for hosting the challenge and releasing the dataset.

Funding.

Federal Ministry of Research, Technology and Space — "KI-Servicezentrum Berlin-Brandenburg" (16IS22092). Responsibility for the content remains with the authors.

Contact.

babajide.owoyele@hpi.de · github.com/babowoyele/seizure-detection